

# ANDREW (KOHLER) WOOD III

## MECHANICAL ENGINEER, EIT

Entry-level mechanical engineer with hands-on experience taking CAD designs from concept through prototype and fabrication.

### CONTACT

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📍 Lakewood, CO

### CERTIFICATIONS

**Engineer in Training EIT**  
Received March 2026

**FE Mechanical Exam**  
Passed October 2025

**Six Sigma Yellow Belt**  
Awarded by Colorado School of Mines  
May 2025

### SKILLS

**CAD**  
SolidWorks parts/assemblies/drawings,  
GD&T basic, tolerance stack-ups basic,  
DFM/DFA

**Simulation**  
FEA linear static, buckling, CFD SolidWorks  
Flow Simulation—basic internal flow/heat  
transfer

**Manufacturing**  
CNC router/mill/laser/waterjet/plasma, MIG  
welding, metal/wood fabrication, FDM/SLA  
3D printing

**Methods/Tools**  
FMEA, Scrum sprints, backlog, standups

**Programming**  
MATLAB, Python, C++, Arduino, LaTeX

### EXTRACURRICULAR & INTERESTS

ASME member · Woodworking · Certified  
Rescue Diver · Advanced skier · Jewelry  
fabrication

### EXPERIENCE

**Prototyping & Fabrication Projects** March 2025 – Present  
LABRIOLA INNOVATION HUB, MINES • GOLDEN, CO

Designed and fabricated woodworking items cutting boards, rolling pins, record cabinet using SolidWorks models/drawings; executed long miter joints requiring tight fit-up and consistent tolerances.

Operated ShopBot CNC router and shop equipment; selected grain orientation and specified food-safe adhesives/finishes; improved dimensional consistency through refined setups.

**Senior Capstone Project – Adaptive Bass Guitar** Aug 2024 – May 2025  
COLORADO SCHOOL OF MINES • GOLDEN, CO

Redesigned the plucking mechanism to eliminate failure modes from flexible/unstable connections; implemented 3D-printed paddle interfaces to improve reliability in repeated use.

Designed a player stabilization system shoulder strap + hip strap to counteract playing-induced torque and replace a tripod-based neck support, improving usability and reducing setup inconvenience.

Modeled assemblies/mechanisms in SolidWorks and produced a detailed assembly manual.

Ran work in 2-week Scrum sprints; tracked risks/mitigations with FMEA.

**Mechanical Design Assistant** Summer 2022  
JG WORKS • HAILEY, ID

Fabricated mono-ski hardware for Special Olympics athletes.

Assisted in designing modular office space using 3-axis CNC cut plywood sheets.

**Fabrication Intern** Summer 2021  
PHRED'S FABRICATION • KETCHUM, ID

Assisted with layout, fabrication, and installation of steel railings for residential projects.

### EDUCATION

**Colorado School of Mines** Oct 2022 – Present  
B.S. Mechanical Engineering GOLDEN, CO

**Macy-Naszkowski School of Art Music**  
Additional Training: Classical Piano Aug 2021 – Aug 2025  
LAKEWOOD, CO